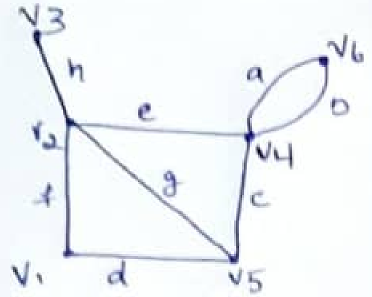
Module-4

Theorem: If edges of a connected graph are arranged in the same order for the columns of the incidence matrix A & the path matrix $P(x,y)$ then the product (mod 2) is $A \cdot P^T(x,y) = M$

where M has 1's in two rows x & y and rest of the $(n-2)$ rows are all 0's

Eg:



*) A is incident matrix

*) P is Path matrix

$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{array} \begin{array}{c} \begin{array}{cccccccc} a & b & c & d & e & f & g & h \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$P(v_3,v_4) = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccccccc} a & b & c & d & e & f & g & h \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

$$P^T(v_3,v_4) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A \cdot P^T(v_3,v_4) = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Mod 2

$0+0=0$
$0+1=1$
$1+0=1$
$1+1=10 \Rightarrow 0$

*) in resultant matrix only v_3 & v_4 rows are having 1's. Remaining are zeros.

$$= \begin{bmatrix} 0 & 0 & 0+0+0+\overset{\text{mod 2}}{1+0+1}+0+0 \\ (1+1) & (1+1) & (1+1) \\ (1+0) & 1 & 1 \\ 1 & 1 & 1 \\ 0 & (1+1)^{\text{mod 2}} & (1+1)^{\text{mod 2}} \\ 0 & 0 & 0 \end{bmatrix} = \begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{array} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Module - 5:

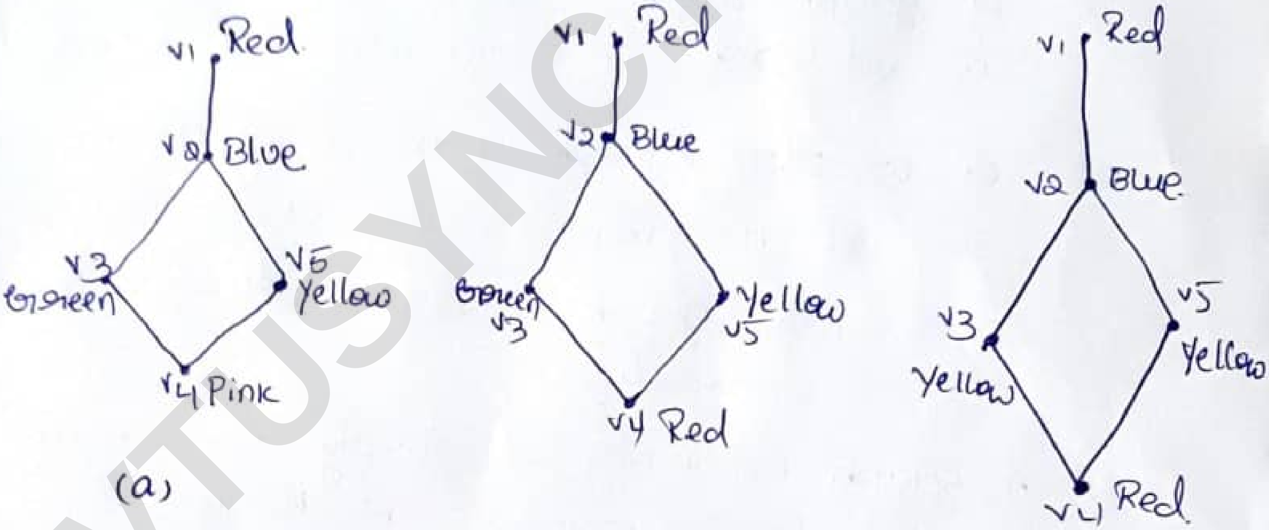
Graph colouring

(colouring graph components) usually we go colouring vertices

* Painting all the vertices of a graph with colors such that no two adjacent vertices have the same color is called proper coloring. (coloring)

* a graph in which every vertex has been assigned a color according to a proper coloring is called a properly colored graph.

* The given graph can be properly colored in many different ways.



* Proper coloring which is of interest to us is one that requires the minimum number of colors.

* a graph G that requires k different colors for its proper coloring & no less is called k-chromatic graph. k → is called chromatic Number.

* for above graph K=3.

note:
* In coloring graphs there is no point in considering disconnected graphs because how we color vertices in one component of a disconnected graph has no effect on the other component

* ∴ it is usually to investigate coloring of connected graph

Some observations that follow directly from definition.

*) (All parallel edges b/n two vertices can be replaced by a single edge without affecting adjacency of vertices

*) self-loops must be disregarded.

*) Thus for coloring problem, we need to consider only simple connected graphs

1. a graph consisting of only isolated vertex. is 1-chromatic.

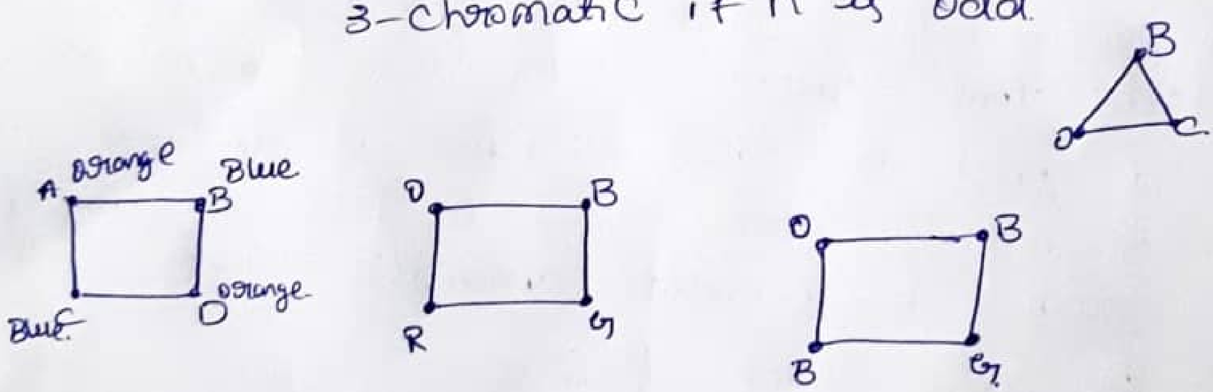
2. a graph with one or more edges (not a self-loop) is at least 2-chromatic (bichromatic.)

3. a complete graph with n vertex is n-chromatic as all its vertices are ajacent.

Eg: every graph having a triangle is atleast 3-chromatic.

4. A graph consisting of simply one circuit with $n \geq 3$ vertices is 2-chromatic if n is even
3-chromatic if n is odd

Eg:



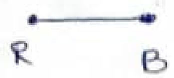
Chromatic Number: minimum no of colors used for proper coloring of graph. denoted by 'K'.

Observation continued.

*) isolated vertex

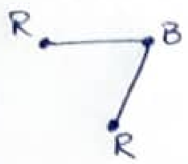
→ 1-chromatic.

*)



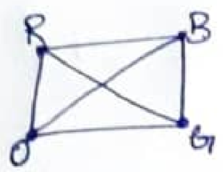
2 colors are required — 2-chromatic.

*)



one or more edge — 2-chromatic

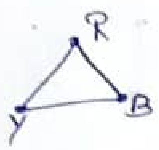
*)



complete graph — 4-chromatic

K_4 — complete graph of 4 vertices

*)



k_3 — 3-chromatic

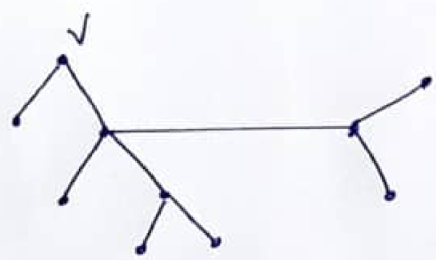
$n \geq 3$ — 3 chromatic
↓ odd

$n = 4$ — 2 chromatic.

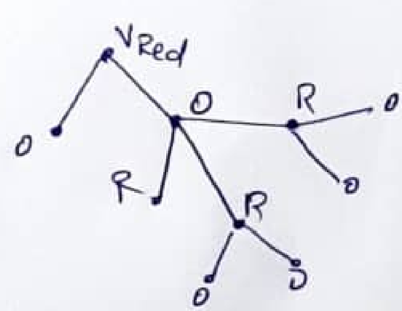
2-chromatic.

Theorem 1

Every tree with two or more vertices is 2-chromatic.

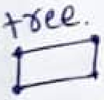


rooted tree



*) Converse of this theorem is not true ie every 2-chromatic graph is tree.

Eg:



*) Tree is minimaly connected.

*) all the vertices which are at odd distence have second color.

*) all the vertices which are at even distence have 1st color

*) every tree with $n \geq 2$ is 2-chromatic

Theorem 2

a graph with atleast one edge is 2 chromatic iff it has no cycles of odd length.

Proof:

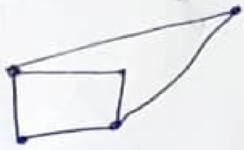
- G - Connected graph with cycles of only even length.
- Spanning tree T of 'G'.

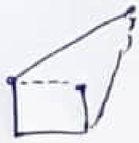
Since tree we can properly color with 2-chromatic.

- add the chords one by one

Since G has no cls of odd length. each chord or each vertex is the chord have different colors

- G is colored with 2 colors





Conversely

if G has cycles of odd length we need atleast 3 colors for proper coloring

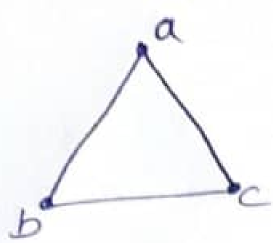
$\therefore$ G cannot be a 2 chromatic.

Theorem 3:

If d_{max} is the maximum degree of the vertices in a graph G then chromatic number of G is

Chromatic number of $G \leq 1 + d_{max}$

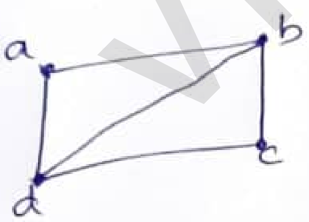
Eg:

1) 

$d(a) = 2$
$d(b) = 2$
$d(c) = 2$

$\therefore$ chromatic no of $G \leq 1 + d_{max}$
$\leq 1 + 2$

Chromatic no is 3 or less than 3
but chromatic no is 3 gives proper coloring

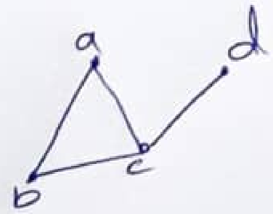
2) 

$d(a) = 2$
$d(b) = 3$
$d(c) = 2$
$d(d) = 3$

$d_{max} = 3$

Chromatic no is 3

$\therefore$ chromatic no $\leq 1 + d_{max}$
$\leq 1 + 3$
≤ 4

3) 

$d(a) = 2$
$d(b) = 2$
$d(c) = 3$

$d_{max} = 3$

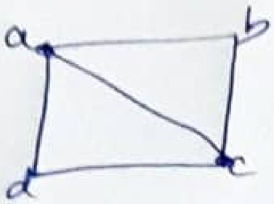
$\therefore$ chromatic no $\leq 1 + d_{max}$
≤ 4

Chromatic no is 3

Theorem3 Continued

if G has no complete graph In that case

Chromatic Number of $G \leq d_{max}$



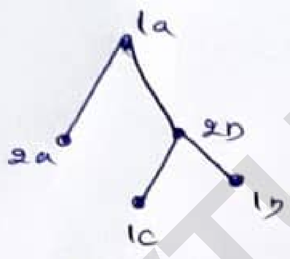
its not a complete graph so

Chromatic Number of $G \leq d_{max}$

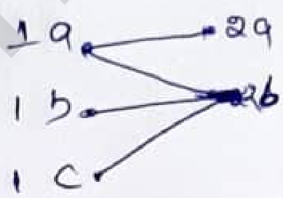
≤ 3

*) A graph G is called bipartite if its vertex set V can be decomposed into 2 disjoint subsets V_1 & V_2 such that every edge in G joins a vertex V_1 with a vertex in V_2.

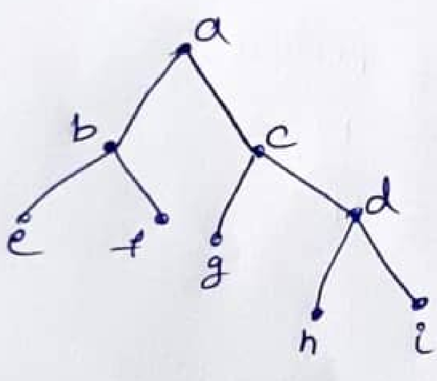
*) Bipartite graph can not have self-loop

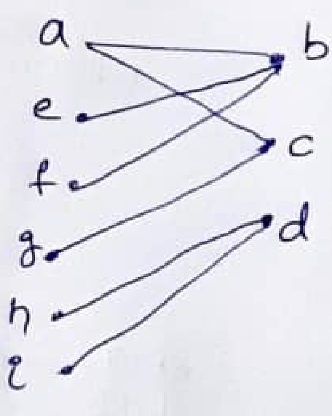


Tree



bipartite





$d_{max} = 3$

Chromatic no is $\leq d_{max}$.

≤ 3

Chromatic polynomial :-

*) The given graph G of 'n' vertices can be proper colored in many different ways using sufficiently many number of colors

*) This property of a graph can be expressed by using a polynomial that polynomial is called Chromatic Polynomial.

*) Chromatic Polynomial denoted by $P_n(\lambda)$. $\lambda \to$ Number of colors
n- Number of vertices graph has.

*) The value of chromatic polynomial of a graph G with 'n' vertices gives me No of different ways of properly coloring the graphs. using λ (lambda) or fewer λ colors.

Expression of λ:

Suppose we have to choose 'i' colors out of λ colors then we write

$$\binom{\lambda}{i} = {}^{\lambda}C_i = \frac{\lambda!}{(\lambda-i)!\, i!}$$

$\because {}^nC_r = \frac{n!}{(n-r)!}$

${}^nC_r = \frac{n!}{(n-r)!\, r!}$

*) Let c_i be the different ways of properly coloring G using 'i' different colors

$\therefore$ G can be properly colored using 'i' colors in $c_i \binom{\lambda}{i}$ ways.

i is +ve integer varies from 1 to n

n is No vertices in G

$$P_n(\lambda) = \sum_{i=1}^{n} c_i \cdot \binom{\lambda}{i}$$

$$= c_1\binom{\lambda}{1} + c_2\binom{\lambda}{2} + c_3\binom{\lambda}{3} + \cdots + c_n\binom{\lambda}{n}$$

$$P_n(\lambda) = c_1 \frac{\lambda!}{(\lambda-1)!\,1!} + c_2 \cdot \frac{\lambda!}{(\lambda-2)!\,2!} + \cdots + c_n \frac{\lambda!}{(\lambda-n)!\,n!}$$

$$= c_1 \lambda + c_2 \frac{\lambda(\lambda-1)}{2} + \cdots + c_n \frac{\lambda(\lambda-1)\cdots(\lambda-n+1)}{n!}$$

Eg:

1) 

(G)

*) 2 minimum color needed to properly coloring

*) c_i – No ways graph can be colored with given i color

$i = 2$.

$c_1 = 0$

2) 

(G)

*) 3 minimum colors needed to properly color the grap G

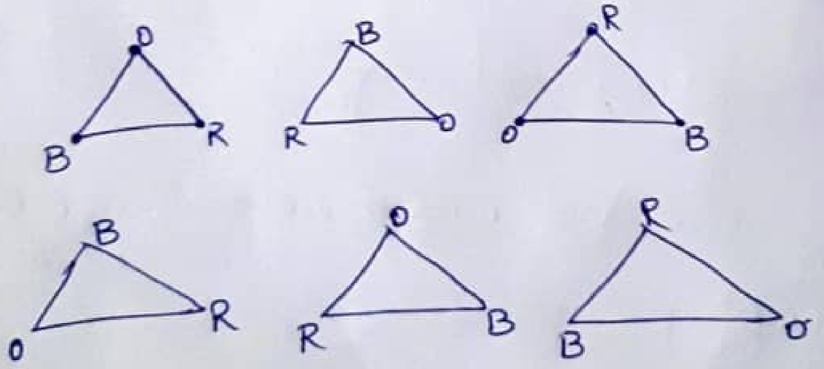
$i = 3$

$\therefore c_1 = 0 \quad c_2 = 0 \quad c_3 = 3!$
$= 3 \times 2 \times 1$
$= 6$

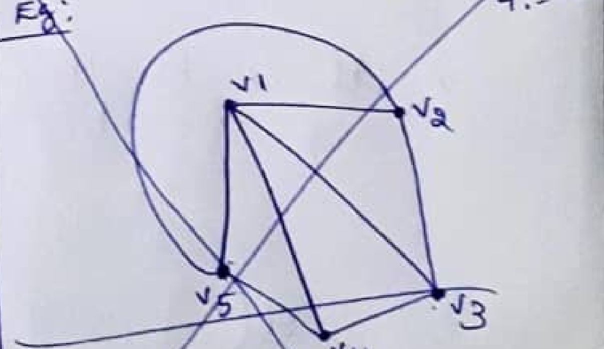
*) graph with n vertices using n different colors we can properly color the graph in $n!$ ways

$c_n = n!$

$c_3 = 3! = 6$



Eg:



3-chromatic graph

*) graph has triangle it will require atleast 3 diffent colors

$c_1 = c_2 = 0$ & $c_5 = n!$

$c_3 = 6$ $\quad 4 \cdot 6 = 28 \quad = 5 \times 4 \times 3 \times 2 \times 1$

$c_4 = 24 \quad = 120$

Theorem 1: A graph of n vertices is a complete graph iff its Chromatic Polynomial is

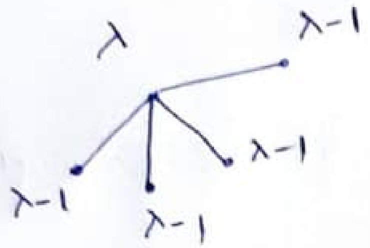
$$P_n(\lambda) = \lambda(\lambda-1)(\lambda-2)\ ----\ (\lambda-n+1)$$

Theorem 2:

An n-vertex graph is a tree iff its chromatic Polynomial $P_n(\lambda) = \lambda(\lambda-1)^{n-1}$

Eg:

1)

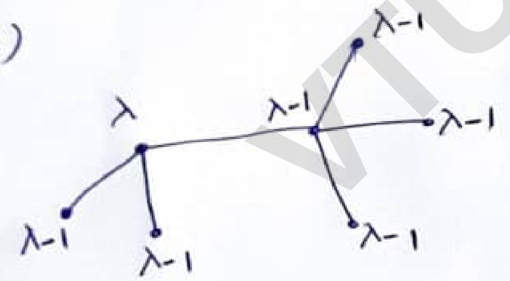


for tree

$$P_n(\lambda) = \lambda(\lambda-1)^{n-1}$$

$$P_5(\lambda) = \lambda(\lambda-1)^{5-1} = \lambda(\lambda-1)^4.$$

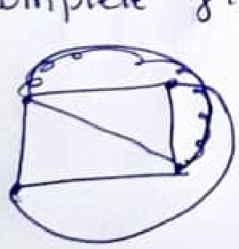
2)



$$P_n(\lambda) = \lambda(\lambda-1)^6.$$

$$= \lambda(\lambda-1)^{7-1} = \lambda(\lambda-1)^6$$

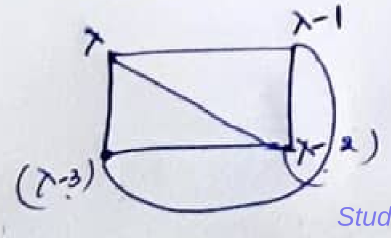
3) complete graph



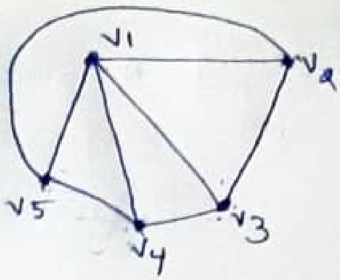
formula for Complete graph

$$P_n(\lambda) = \lambda(\lambda-1)(\lambda-2)\ ----\ (\lambda-n+1)$$

$$= \lambda(\lambda-1)(\lambda-2)(\lambda-3)$$



Eg:



$c_1 = 0 \quad c_2 = 0 \qquad c_3 = 3!$

$= 3 \times 2 \times 1$

$= \lambda(\lambda-1)(\lambda-2)$

$c_4 = \cancel{\lambda}$

$= 4 \times 3 \times 2 \times 1 = 24 \cdot 2.$

$= \lambda(\lambda-1)(\lambda-2)(\lambda-3) \cdot 2$ — two news are possible

$c_5 = \lambda(\lambda-1)(\lambda-2)(\lambda-3)(\lambda-4)$

$P_5(\lambda) = c_3 + c_4 + c_5$

$= \lambda(\lambda-1)(\lambda-2) + 2\lambda(\lambda-1)(\lambda-2)(\lambda-3) + \lambda(\lambda-1)(\lambda-2)(\lambda-3)(\lambda-4)$

$= \lambda(\lambda-1)(\lambda-2)(\lambda^2 - 5\lambda + 7)$

$P_5(\lambda) = \lambda(\lambda-1)(\lambda-2)(\lambda^2 - 5\lambda + 7)$

$\lambda = 5 \quad = 5(4)(3)(5^2 - 5 \times 5 + 7)$

$= 60 \cdot (\cancel{25} - \cancel{25} + 7)$

$= 480$

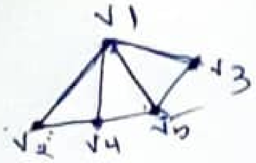
$P_5(\lambda) = c_1\binom{\lambda}{1} + c_2\binom{\lambda}{2} + c_3\binom{\lambda}{3} + c_4\binom{\lambda}{4} + c_5\binom{\lambda}{5}$

$c_1 = 0 \quad c_2 = 0$

$= c_3\binom{\lambda}{3} + c_4\binom{\lambda}{4} + c_5\binom{\lambda}{5}$

$c_3 = \cancel{6}\,\dfrac{\lambda!}{(\lambda-3)!\cancel{3!}} + \cancel{4!} \cdot 2^{(1+1)} \dfrac{\lambda!}{(\lambda-4)!\cancel{4!}} + \cancel{5!}\,\dfrac{\lambda!}{(\lambda-5)!\cancel{5!}}$

$= \dfrac{\lambda!}{(\lambda-3)!} + 2\dfrac{\lambda!}{(\lambda-4)!} + \dfrac{\lambda!}{(\lambda-5)!} = \lambda(\lambda-1)(\lambda-2) + 2\lambda(\lambda-1)(\lambda-2)(\lambda-3) + \lambda(\lambda-1)(\lambda-2)(\lambda-3)(\lambda-4)$

Eg. 

$$P_n(\lambda) = c_1\binom{\lambda}{1} + c_2\binom{\lambda}{2} + c_3\binom{\lambda}{3} + \cdots + c_n\binom{\lambda}{n}$$

$$P_5(\lambda) = c_1\binom{\lambda}{1} + c_2\binom{\lambda}{2} + c_3\binom{\lambda}{3} + c_4\binom{\lambda}{4} + c_5\binom{\lambda}{5}$$

$$= c_1\frac{\lambda!}{(\lambda-1)!\,1!} + c_2\frac{\lambda!}{(\lambda-2)!\,2!} + c_3\frac{\lambda!}{(\lambda-3)!\,3!} + c_4\frac{\lambda}{(\lambda-4)!\,4!} + c_5\frac{\lambda}{(\lambda-5)!\,5!}$$

or

$$= c_1\frac{\lambda}{1!} + c_2\frac{\lambda(\lambda-1)}{2!} + c_3\frac{\lambda(\lambda-1)(\lambda-2)}{3!} + c_4\frac{\lambda(\lambda-1)(\lambda-2)(\lambda-3)}{4!}$$

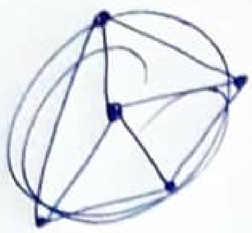
$$+ c_5\frac{\lambda(\lambda-1)(\lambda-2)(\lambda-3)(\lambda-4)}{5!}$$

$c_1 = 0$, $c_2 = 0$, $c_5 = 5!$

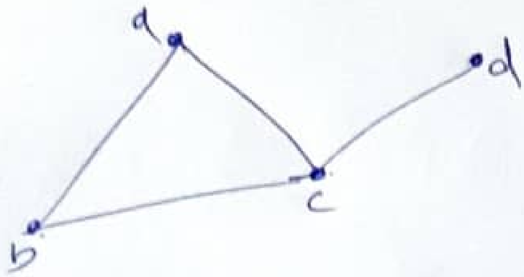
$c_3 = 3!$ $\qquad$ $c_4 = 4!\cdot(2+1)$

4×3×2
24.

$$P_5(\lambda) = \lambda(\lambda-1)(\lambda-2)(\lambda^2-4\lambda+4)$$



(G)



3 chromatic

$c_1 = 0 \qquad c_2 = 0$

$c_3 = 3! \cdot 1 = 3!$

$c_4 =$ ∵ $c_n = n!$

$\therefore c_4 = 4!$

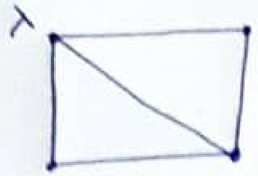
$$P_4(\lambda) = \cancel{3!}\,\frac{\lambda(\lambda-1)(\lambda-2)}{\cancel{3!}} + \cancel{4!}\,\frac{\lambda(\lambda-1)(\lambda-2)(\lambda-3)}{\cancel{4!}}$$

$$= \lambda(\lambda-1)(\lambda-2) + \lambda(\lambda-1)(\lambda-2)(\lambda-3)$$

$$= \lambda(\lambda-1)(\lambda-2)(1 + \lambda-3)$$

$$= \lambda(\lambda-1)(\lambda-2)(\lambda-2)$$

$$P_4(\lambda) = \lambda(\lambda-1)(\lambda-2)^2$$

4) 

formula

$$P_n(\lambda) = \sum_{i=1}^{n} c_i \binom{\lambda}{i} = \sum_{i=1}^{4} c_i \binom{\lambda}{i}$$

$$= c_1 \binom{\lambda}{1} + c_2 \binom{\lambda}{2} + c_3 \binom{\lambda}{3} + c_4 \binom{\lambda}{4}$$

$$= c_1 \frac{\lambda}{(\lambda-1)!} + c_2 \frac{\lambda!}{(\lambda-2)!\,2!} + c_3 \frac{\lambda!}{(\lambda-3)!\,3!} + c_4 \frac{\lambda!}{(\lambda-4)!\,4!}$$

Here $c_1 = 0$, $c_2 = 0$, $c_4 = 4!$

we have to find c_3

$$c_3 = 3! \cdot 1 = 6 \cdot 1 = 6.$$

$$P_4(\lambda) = 0 + 0 + \cancel{6}\, \frac{\lambda!}{(\lambda-3)!\,\cancel{3!}} + \cancel{4!}\, \frac{\lambda!}{(\lambda-4)!\,\cancel{4!}}$$

$$= \frac{\lambda!}{(\lambda-3)!} + \frac{\lambda!}{(\lambda-4)!}$$

$$= \lambda(\lambda-1)(\lambda-2) + \lambda(\lambda-1)(\lambda-2)(\lambda-3)$$

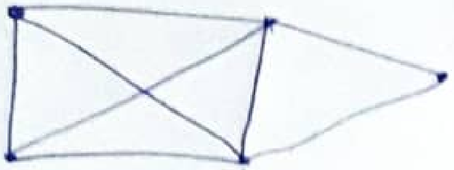
$$= \lambda(\lambda-1)(\lambda-2)^2.$$

$$n! = n \times (n-1) \times (n-2) \times (n-3) \cdots (n-n+1)$$

$$(\lambda-3)! = (\lambda-3)(\lambda-3-1)(\lambda-3-2) \cdot (\lambda-3-(\lambda-3)+1)$$

$$= (\lambda-3)(\lambda-4)(\lambda-5)$$

Eg:



Chromatic polynomial is 4.

$\therefore\ c_1 = c_2 = c_3 = 0$

$c_4 = 4! \cdot 2 = 4 \times 3 \times 2 \cdot 2 = 24 \cdot 2 = 48$

$c_5 = 5!$

$$P_5(\lambda) = c_3\binom{\lambda}{3} + c_4\binom{\lambda}{4} + c_5\binom{\lambda}{5}$$

$$= 0 + c_4 \frac{\lambda(\lambda-1)(\lambda-2)(\lambda-3)}{4!} + c_5 \frac{\lambda(\lambda-1)(\lambda-3)(\lambda-4)}{5!}$$

$$= \cancel{4!}\, 2\, \frac{\lambda(\lambda-1)(\lambda-2)(\lambda-3)}{\cancel{4!}} + \cancel{5!}\, \frac{\lambda(\lambda-1)(\lambda-3)(\lambda-4)}{\cancel{5!}}$$

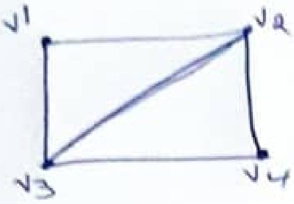
$$= 2\lambda(\lambda-1)(\lambda-2)(\lambda-3)\left[1 + (\lambda-4)\right]$$

$$= 2\lambda(\lambda-1)(\lambda-2)(\lambda-3)(\lambda-3)$$

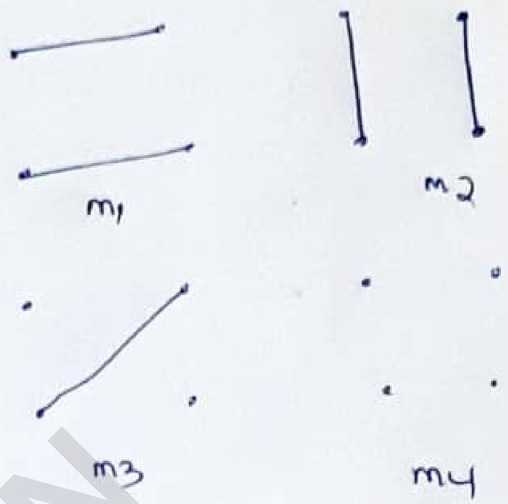
$$P_5(\lambda) = 2\lambda(\lambda-1)(\lambda-2)(\lambda-3)^2$$

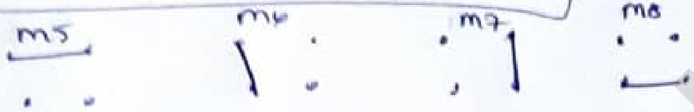
Matching:

Matching in a graph is a subset of edges in which no 2 edges are adjacent.

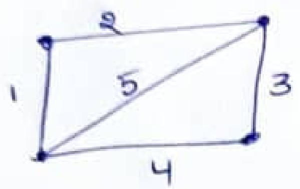


(v_1, v_2)
(v_2, v_4)
(v_3, v_4)
(v_1, v_4)
(v_2, v_3)





Adjacent edges:- edges that share a common vertex.



(G)

1, 2 are adjecent
2, 3 —〃—
3, 4 —〃—
1, 4 —〃—

Matching:- for graph G.

$\{5\}$
$\{2, 4\}$
$\{1, 3\}$

Matching Number:

*) No of edges in a largest maximal matching is called matching Number in this eg it's 2.

Maximal matching: is a matching to which no edge in the graph can be added.

m_1, m_2 & m_3 are maximal matching

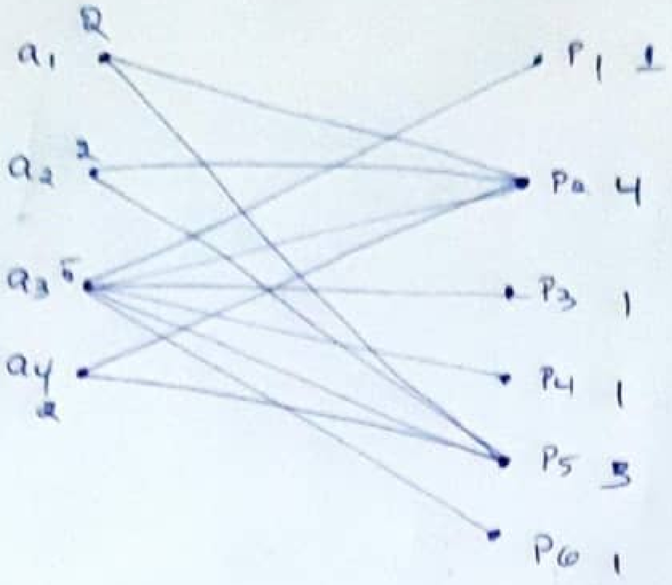
Maximum matching

m_1 & m_2

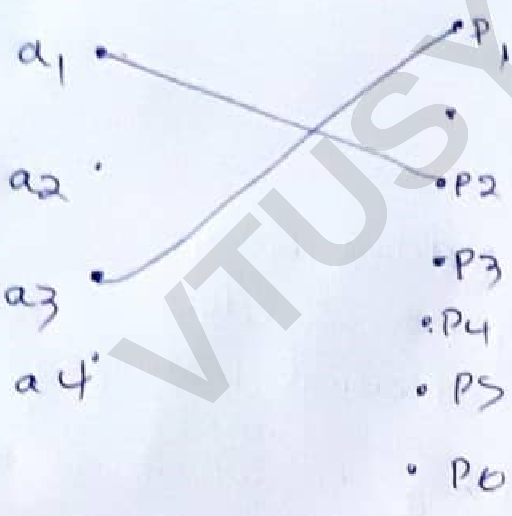
Perfect matching:

all the vertex of the matching is 1

m_1 & m_2 Perfect matching



$\{a_1, a_2, a_4\}$ have edges to P_2 & P_5



Complete matching:

In bipartite graph, vertex partition V_1 & V_2
a complete matching of vertices in set V_1 into those V_2
is a matching in which there is one edge incident
with every vertex in V_1.

or

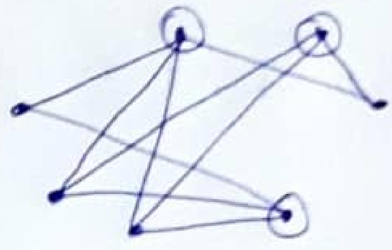
V_1 is matched against some vertex in V_2.

Bipartite graph:

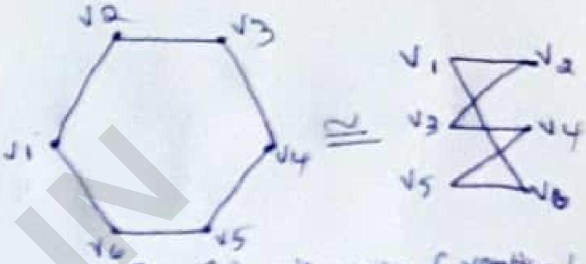
1) Basically Bipartite graph is 2-chromatic

* An equivalent definition of a bipartite graph is that graph where it is possible to color the vertices red or blue so that no 2 adjacent vertices are the same color.

*) Eg:



$K_{4,3}$



C_6 (circuit with 6 vertices) is also a bipartite 2-chromatic

bipartite graph but not complete.

Complete matching:-

In bipartite graph $G=(V, E)$ with bipartition (V_1, V_2) a matching M is said to be a complete matching form V_1 to V_2 iff $|M| = |V_1|$ every vertex in V_1 is end point of an edge in M.

Size of the matching = Size of the vertex in V

= 5

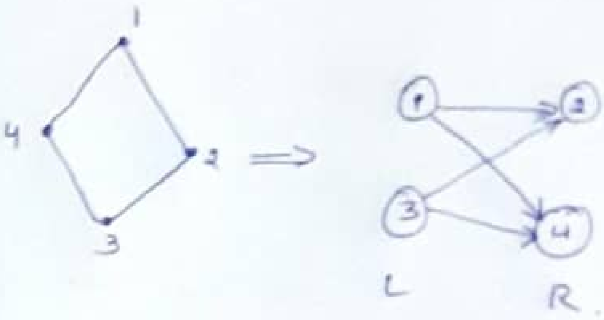
Complete bipartite graph

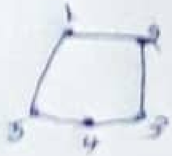
K_{mn} no edges $|E| = m.n$

Eg: Bipartite graphs are used to model applications that involve matching the elements of one set to elements in another

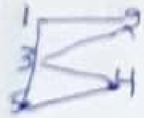
Eg: Job assignment - vertices represents jobs and employees edges links employee with jobs they have been trained with goal is match jobs to employees so most jobs are done

Eg:





$K_{2,2}$



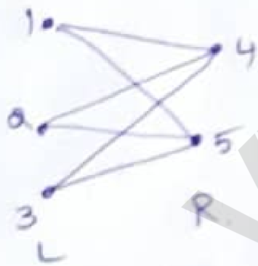
complete bipartite graph.

*) C_5 — can't be bipartite graph

any cycle with odd length can't be converted into bipartite graph

Maximum Matching in bipartite graph

1)



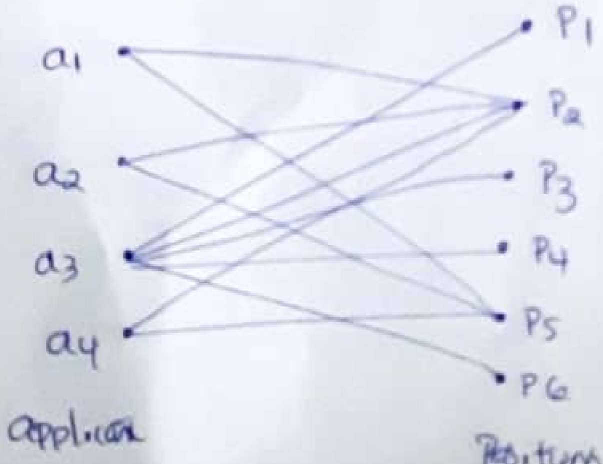
maximum matching

(1, 4, 2, 5) matched vertex.

3 unmatched vertex

matching length or matching Number is 2

2)



a_1 is Qualified for P_2 & P_5

a_2 is Qualified for P_2 & a_5.

a_4 is Qualified for P_2 & P_1

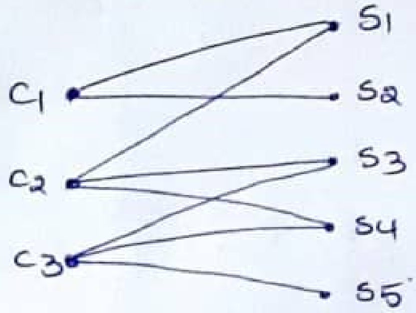
a_3 is Qualified for P_1, P_2, P_3, P_4 ...

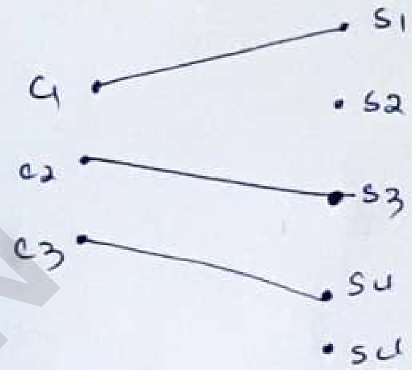
Matching:-

Theorems:

a complete Matching of V_1 into V_2 in a bipartite graph exists iff every subset of 'r' vertices in V_1 is collectively adjacent to 'r' or more vertices in V_2.

Eg:





M.

$r = 1$ $\{c_1\}$ $\{s_1, s_2\}$

$\{c_2\}$ $\{s_1, s_3, s_4\}$

$\{c_3\}$ $\{s_3, s_4, s_5\}$

$r = 2$ $\{c_1, c_2\}$ $\{s_1, s_2, s_3, s_4\}$

$\{c_2, c_3\}$ $\{s_1, s_3, s_4, s_5\}$

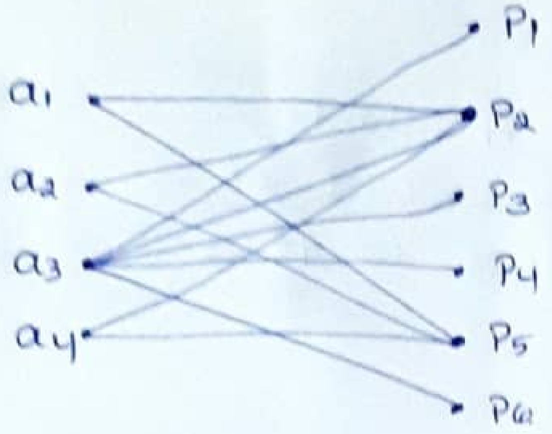
$\{c_3, c_1\}$ $\{s_1, s_2, s_3, s_4, s_5\}$

$r = 3$ $\{c_1, c_2, c_3\}$ $\{s_1, s_2, s_3, s_4, s_5\}$

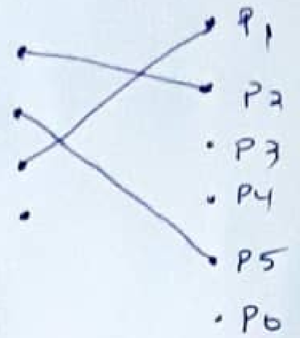
Eg: 2

Prove that the given graph does not have complete Matching.

$4C_2 = \frac{4!}{(4-2)!\,2!} = \frac{4!}{2!\,2!} = \frac{4\times3\times2}{2\times1\times2\times1} = 6$

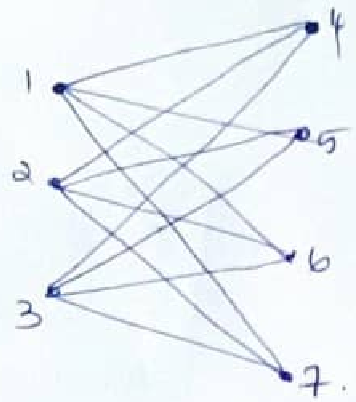


$4C_3 = \frac{4!}{(4-3)!\,3!} = \frac{4\times3\times2}{1\cdot3\times2} = 4$

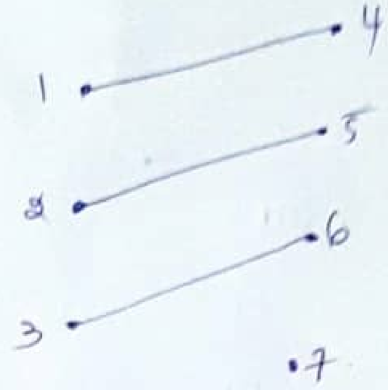


	V_1	V_2
$r=1$	$\{a_1\}$	$\{P_2, P_5\}$
	$\{a_2\}$	$\{P_2, P_5\}$
	$\{a_3\}$	$\{P_1, P_2, P_3, P_4, P_6\}$
	$\{a_4\}$	$\{P_2, P_5\}$
$r=2$	$\{a_1, a_2\}$	$\{P_2, P_5\}$
	$\{a_2, a_3\}$	$\{P_1, P_2, P_3, P_4, P_5, P_6\}$
	$\{a_3, a_4\}$	$\{P_1, P_2, P_3, P_4, P_5, P_6\}$
	$\{a_1, a_4\}$	$\{P_2, P_5\}$
	$\{a_1, a_3\}$	$\{P_1, P_2, P_3, P_4, P_5, P_6\}$
	$\{a_2, a_4\}$	$\{P_2, P_5\}$
$r=3$	$\{a_1, a_2, a_3\}$	$\{P_1, P_2, P_3, P_4, P_5, P_6\}$
	$\{a_2, a_3, a_4\}$	$\{P_1, P_2, P_3, P_4, P_5, P_6\}$
	$\{a_3, a_4, a_1\}$	$\{P_1, P_2, P_3, P_4, P_5, P_6\}$
	$\boxed{\{a_4, a_1, a_2\}}$ 3	$\boxed{\{P_2, P_5\}}$
$r=4$	$\{a_1, a_2, a_3, a_4\}$	$\{P_1, P_2, P_3, P_4, P_5, P_6\}$

Complete bipartite diagram



K 3.4



Complete matching

r_1	1	{4, 5, 6, 7}
	2	{4, 5, 6, 7}
	3	{4, 5, 6, 7}
r_2	(1, 2)	{4, 5, 6, 7}
	(2, 3)	{4, 5, 6, 7}
	(3, 1)	{4, 5, 6, 7}
	(1, 4)	{4, 5, 6, 7}
	(1, 3)	{4, 5, 6, 7}
	(2, 4)	{4, 5, 6, 7}
		{4, 5, 6, 7}
r_3	{1, 2, 3}	{4, 5, 6, 7}
	~~{2, 3, 4}~~	~~{4, 5, 6, 7}~~
	~~{3, 4, 1}~~	~~{4, 5, 6, 7}~~
	~~{4, 1, 2}~~	~~{4, 5, 6, 7}~~

Eg: 3 Three boys b_1, b_2, b_3 & 4 girls g_1, g_2, g_3, g_4 are set

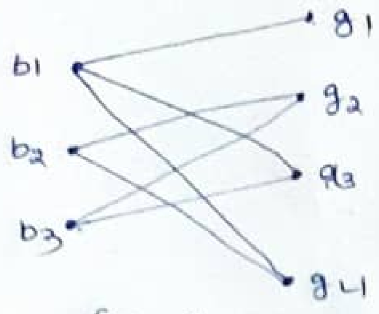
i) b_1 is a cousin of g_1, g_3, g_4.

ii) b_2 is a cousin of g_2 & g_4.

iii) b_3 is a cousin of g_2 & g_3.

Can every one of the boys marry a girl who is one of his cousins? if so find 5 possible sets of such couple.

Hint: Here we have to find whether a complete matching exist from V_1 to V_2.



(G1)

degree of $v_1 \geq 2 \geq$ degree of v_1

3 or more 2 less

	v_1	v_2
r_1	b1	{g1, g3, g4}
	b2	{g2, g4}
	b3	{g2, g4}
r_2	b1, b2	{g1, g2, g3, g4}
	b2, b3	{g2, g3}
	b3, b1	{g1, g2, g3, g4}
$r=3$	{b1, b2, b3}	{g1, g2, g3, g4}

*) for each v_1 no elements in v_2 is greater than or equal to the no of elements in v_1

$\therefore$ the graph has a complete matching.

*) each boy can marry a girl who is one of his cousins

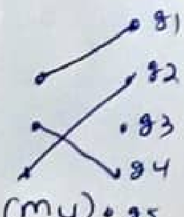
(m1) 1) (b1, g1) (b2, g2) (b3, g3)

(m2) 2) (b1, g3) (b2 g4) (b3, g2)

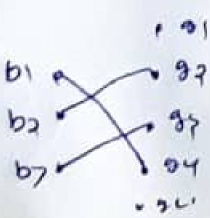
(m3) 3) (b1, g1) (b2, g4) (b3, g2)

(m4) 4) (b1, g1) (b2, g4) (b3, g2)

(m5) 5) (b1, g4) (b2, g2) (b3, g3)



(m4)



(m5)

***

Theorem :-

In a bipartite graph a complete matching of V_1 into V_2 exists if there exists a +ve integer m for which the following condition is satisfied.

degree of every vertex $\geq m$ V_1

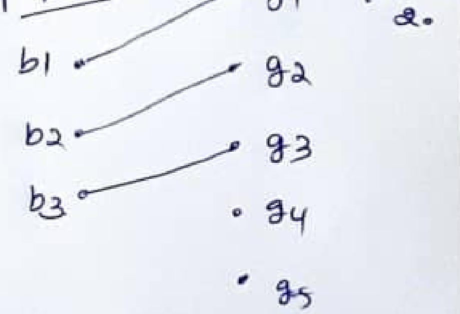
$m \geq$ degree of vertex of V_2

*) in given graph G1

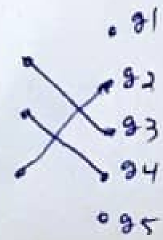
$m = 2$.

*) in V_1 set every vertex degree is 3 ie greater than 2

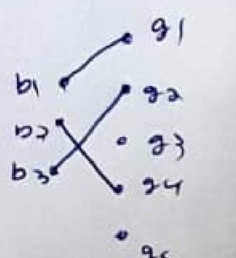
*) in V_2 every vertex degree is 2 ie less than 2.



M_1

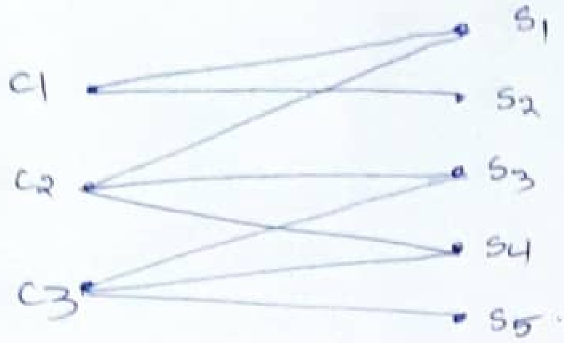


m2



m3

Eg: Complete matching

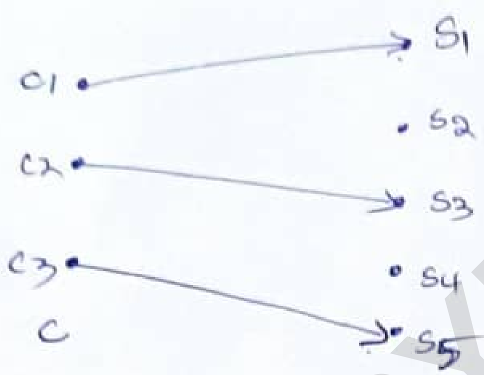


c_1 to $\{s_1, s_2\}$

c_2 to $\{s_1, s_3, s_4\}$

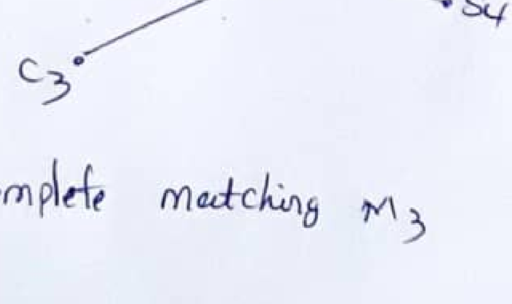
c_3 to $\{s_3, s_4, s_5\}$

Complete matching



Complete matching M_1

Complete matching M_2



Complete matching M_3

Matrix representation of ~~com~~ bipartite graph

$$\begin{array}{c} \\ c_1 \\ c_2 \\ c_3 \end{array} \begin{array}{c} \begin{array}{ccccc} s_1 & s_2 & s_3 & s_4 & s_5 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

$n_1 = 3, \quad n_2 = 5, \quad n = 8$

$n_1 \leq n_2$

$V_1 = \{c_1, c_2, c_3\}$

$V_2 = \{s_1, s_2, s_3, s_4, s_5\}$

Complete matching of V_1 to V_2

$$M = \begin{array}{c} \\ c_1 \\ c_2 \\ c_3 \end{array} \begin{array}{c} \begin{array}{ccccc} s_1 & s_2 & s_3 & s_4 & s_5 \end{array} \\ \begin{vmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{vmatrix} \end{array}$$

Coverings:-

*) In a graph G, a set g of edges is said to cover 'G' if every vertex is incident on at least one edge in g.

*) A spanning tree in a connected graph is a covering.

*) A Hamiltonian circuit (if it exists) in a graph is also a covering.

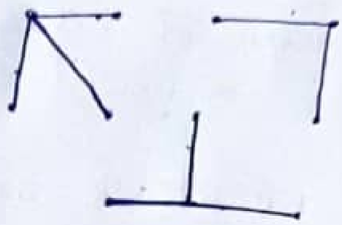
*) a graph G is trivially its own covering.

minimal covering:

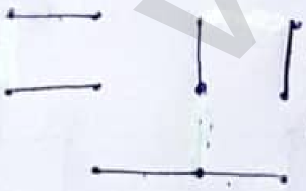
a covering from which no edge can be removed without destroying its ability to cover the graph.

Eg:

(a)

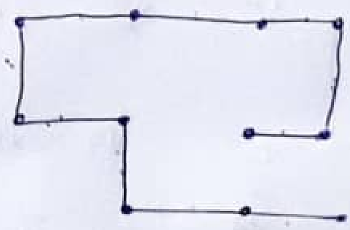


covering 1 (8 edges)



(edges 6)

Covering 2



(10 edges)

spanning tree

Covering Number is 6

Some observation

1. covering exists for a graph if and only graph has no isolated vertex.

2. covering of an n-vertex graph will have at least $\lceil n/2 \rceil$ edges ($\lceil n \rceil$ denotes the smallest integer not less than x)

3. every pendant edge in a graph is included in every covering of the graph

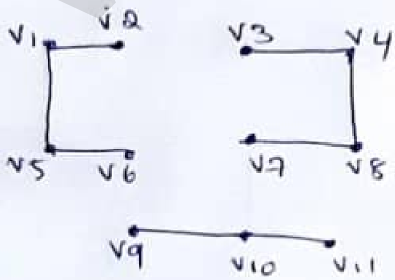
4. every covering contain minimal coverings

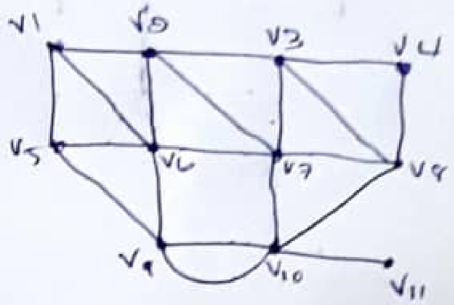
5. no minimal covering can contain a circuit

6. A graph in general has many minimal coverings and that may be of different sizes (number of edges)

7. Number of edges in a minimal covering of the smallest size is called covering number of the graph.

8. If we denote the remaining edges of a graph by $(G-g)$, set of edges g is covering iff every vertex v the degree of vertex in $(G-g) \leq$ (degree of vertex in G) -1





$v_1 = 2$ $v_7 = 1$
$v_2 = 1$ $v_8 = 2$
$v_3 = 1$ $v_9 = 1$
$v_4 = 2$ $v_{10} = 2$
$v_5 = 2$ $v_{11} = 1$
$v_6 = 1$

$v_1 = 3$ $v_5 = 3$
$v_2 = 3$ $v_6 = 5$ $v_{11} = 1$
$v_3 = 3$ $v_7 = 5$
$v_4 = 4$ $v_8 = 4$
$v_9 = 4$
$v_{10} = 5$

$\frac{11}{2} = 5.5 = 6$

* a covering g of a graph is minimal if & only iff g contains no paths of length 3 or more.

Four color problem:

*) So far we have considered proper coloring of vertices
*) let us briefly consider the proper coloring of regions in a planar graph (embedded on a plane or sphere)

*) Just as in coloring of vertices the regions of a planar graph are said to be properly colored if no two contiguous or adjacent regions have the same color.

*) two regions are said to be adjacent if they have a common edge b/n them.
one are more edges are common among the regions of planar graph.

*) we are not interested in just properly coloring the regions of given graph. we are interested in a coloring that uses minimal Number of Colors.

*) This leads us to most famous conjecture in graph theory.

*) 4-color conjecture is worked by many mathematicians

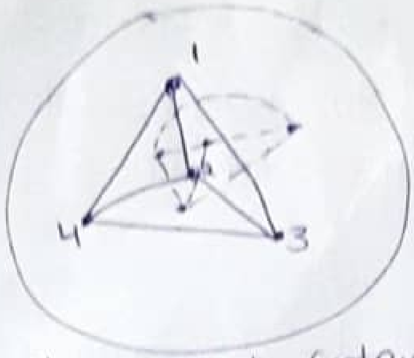
*) To color the maps on the surface of a torus 7 colors are sufficient.

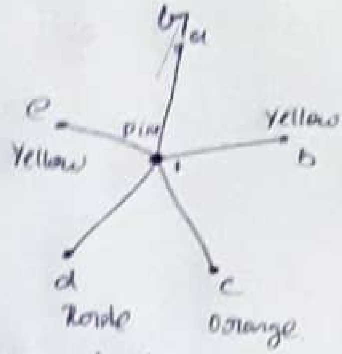
Conjecture → an opinion or conclusion formed on the basis of incomplete information.

5 colour problem:

*) we know that graph has duals if it is planar
∴ coloring the regions of a planar graph G is equivalent to coloring the vertice of its dual, vice versa,

In this case we shall show that every planar map can be properly colored with 5 colors.



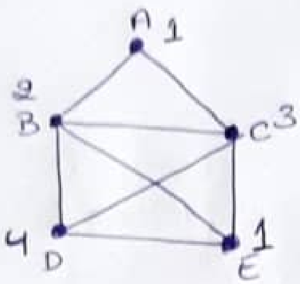


∵ every planar graph contains a vertex with degree $deg(v) \leq 5$

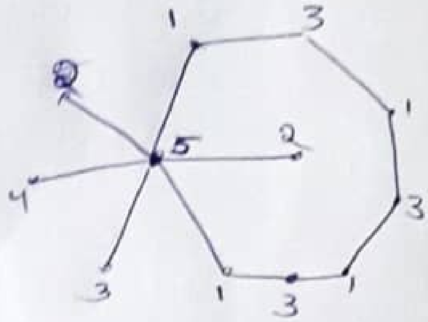
Greedy Algorithm: (Greedy coloring algorithm)

follows local optimal choice at each stage with intend of finding global optimum

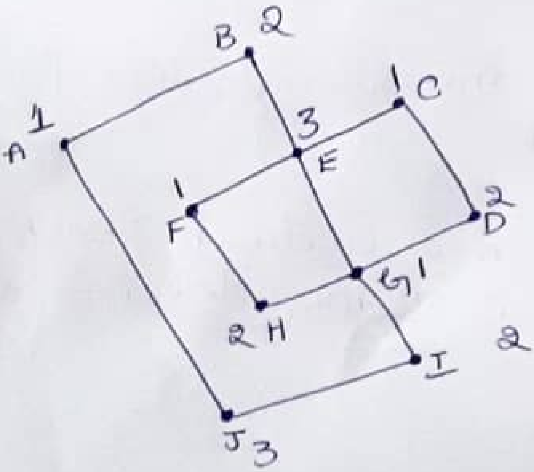
every graph $P(n \leq 5)$



1. Blue
2. green
3. Red
4. Black



(1)



Using greedy colouring algorithm we need 3 colors